

Week 3: Kernel Density Estimation

MATH-517 Statistical Computation and Visualization

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Section 1

Univariate Density Estimation

The problem

Setup: $X_1, \dots, X_n$ is a random sample from a distribution F with continuous density $f(x)$

Goal: Estimate f non-parametrically, i.e., without assuming a particular form

The **histogram**

$$h(x) = \frac{1}{n} \sum_{k=1}^K \frac{\#\left\{\{X_j\}_{j=1}^n \cap \mathcal{I}_k\right\}}{|\mathcal{I}_k|} \mathbb{1}_{x \in \mathcal{I}_k}$$

is the simplest form of density estimation. It requires a specification of

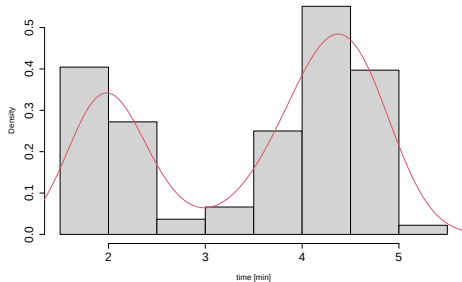
- *origin* and *binwidth*, or
- *breaks*: more general, but non-equidistant binning is bad anyway, so think only about origin and binwidth

Running Ex.: Yellowstone's Old Faithful geyser - faithful data:

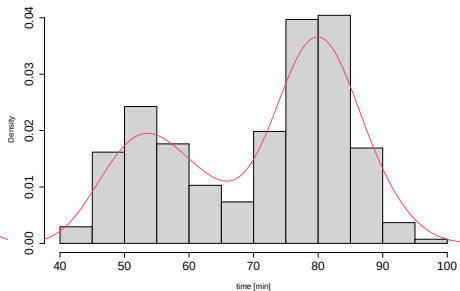
- waiting - time between eruptions
- eruptions - duration of the eruptions

Basic estimator: Histogram

Eruption duration

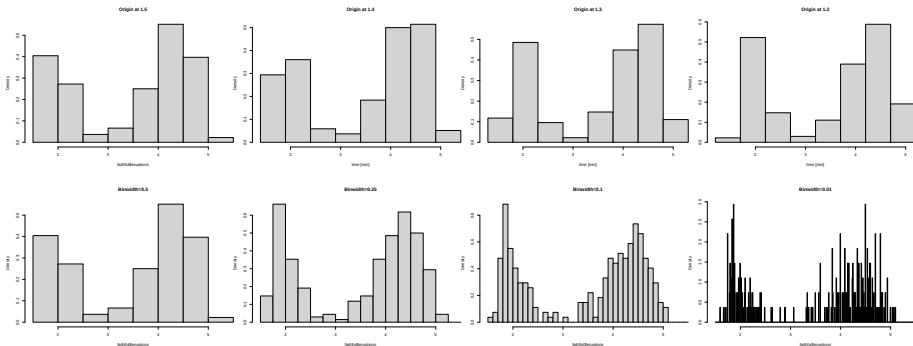


Waiting time



(equally spaced) breaks specified, so a rule of thumb used to choose origin and binwidth

Histogram: Change in Origin and Binwidth

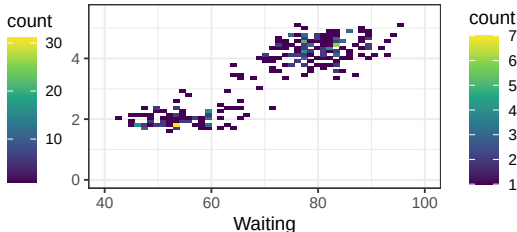
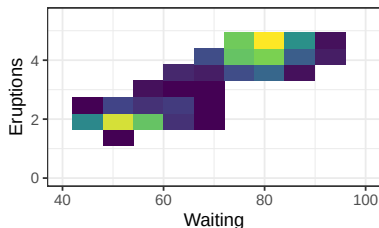


⇒ density estimate depends on the starting position and width of the bins

Issues with Histogram

Histogram is great for visualization, but fails as a density estimator

- *origin* is completely arbitrary
- *binwidth* relates to smoothness of f , but histogram cannot be smooth. The discontinuities of the estimate are not due to the underlying density but to bins' locations and widths
- *curse of dimensionality*: number of bins grows exponentially with the number of dimensions

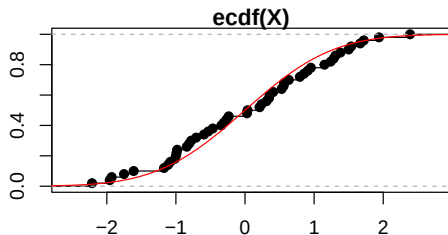
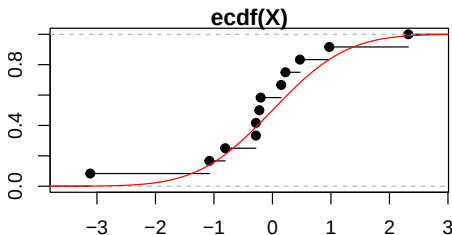


Let us now address these issues by a naive version of kernel density estimation

ECDF

Let $\widehat{F}$ denote the empirical (cumulative) distribution function (ECDF) of the data $\{X_i\}_{i=1}^n$, i.e.,

$$\widehat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{[X_i \leq x]}$$



Naive Density Estimator

- The ECDF $\widehat{F}_n(x)$ is an estimator of F
 - by [Glivenko-Cantelli theorem](#) uniformly almost surely consistent:

$$\sup_x |\widehat{F}_n(x) - F(x)| \xrightarrow{a.s.} 0$$

- Note that f is the derivative of F . However, plugging $\widehat{F}_n(x)$ results in a sum of point masses at the observations as $\widehat{F}_n$ is discrete
- But,

$$f(x) = \lim_{h \rightarrow 0_+} \frac{F(x+h) - F(x-h)}{2h}$$

and we can fix $h = h_n$ small and depending on n , and plug it in:

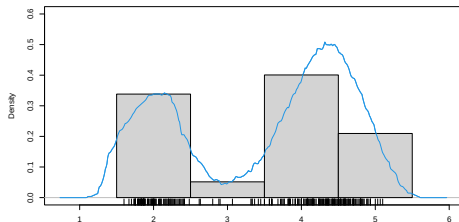
$$\widehat{f}(x) = \widehat{f}_n(x) = \frac{\widehat{F}_n(x+h_n) - \widehat{F}_n(x-h_n)}{2h_n} = \frac{1}{2h_n} \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{[X_i \in (x-h_n, x+h_n)]}$$

⇒ This is called the naive density estimator

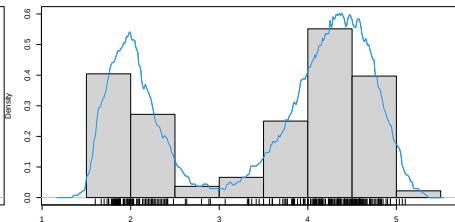
Naive Density Estimator

- The naive DE $\hat{f}$ is a step function with jumps at the points $X_i \pm h$, and thus discontinuous
- $\hat{f}$ is the sum of boxcar functions centered at the observations with width $2h$ and area $1/n \Rightarrow$ this is equivalent to the notion of moving histogram with binwidth= $2h$
 - aggregate data in intervals of the form $(x - h, x + h)$ and approximate the density at x by the relative frequency in $(x - h, x + h)$
 - *origin* does not matter anymore

bandwidth = 0.5, binwidth = 1



bandwidth = 0.25, binwidth = 0.5



Consistency

Theorem If $h = h_n \rightarrow 0$ and $nh_n \rightarrow \infty$ as $n \rightarrow \infty$, then, for any continuity point t ,

$$\hat{f}_n(t) \xrightarrow{p} f(t),$$

as $n \rightarrow \infty$. Thus, $\hat{f}_n$ is a (weakly) consistent estimator

For instance, since

$$\hat{f}(x) = \frac{1}{2nh_n} \sum_{i=1}^n \overbrace{\mathbb{1}_{[X_i \in (x-h_n, x+h_n)]}}^{\text{Ber}(F(x+h_n)-F(x-h_n))}$$

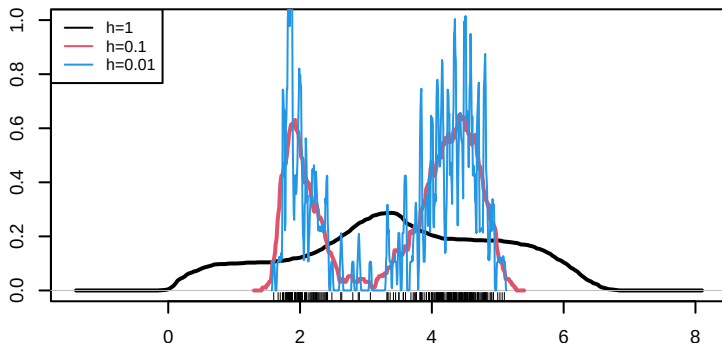
- $\mathbb{E}\hat{f}(x) = \frac{F(x+h_n)-F(x-h_n)}{2h_n} \rightarrow f(x) \quad \text{as } h_n \rightarrow 0_+, \text{ when } n \rightarrow \infty$
- $\text{var}\{\hat{f}(x)\} = \frac{1}{4nh_n^2} \{F(x+h_n) - F(x-h_n)\} \{1 - F(x+h_n) + F(x-h_n)\}$
 $= \frac{F(x+h_n)-F(x-h_n)}{2h_n} \frac{1-F(x+h_n)+F(x-h_n)}{2nh_n} \rightarrow 0$

as $h_n \rightarrow 0$ and $nh_n \rightarrow \infty$, when $n \rightarrow \infty$

Smoothness of the Naive DE

Smoothness of $\hat{f}$ depends on the *bandwidth* h , often called the *smoothing parameter*

- the bandwidth h is a *tuning parameter* and needs to be chosen somehow in practice
 - h small $\rightarrow$ wiggly estimator
 - h large $\rightarrow$ smooth estimator



Naive DE Rewritten

The naive DE can be written as

$$\begin{aligned}\hat{f}(x) &= \frac{1}{2nh_n} \sum_{i=1}^n \mathbb{1}_{[X_i \in (x-h_n, x+h_n)]} = \frac{1}{2nh_n} \sum_{i=1}^n \mathbb{1}_{[-1 < \frac{X_i - x}{h_n} \leq 1]} \\ &= \frac{1}{n} \sum_{i=1}^n \frac{1}{h_n} K\left(\frac{x - X_i}{h_n}\right)\end{aligned}$$

where $K(t) = \frac{1}{2} \mathbb{1}_{\{-1 < t \leq 1\}}$ is the density of $U[-1, 1]$.

- Since $\int_{-\infty}^{+\infty} K(t)dt = 1$, we have that

$$\int_{-\infty}^{+\infty} \hat{f}(x)dx = \frac{1}{n} \sum_{i=1}^n \frac{1}{h_n} \int_{-\infty}^{+\infty} K\left(\frac{x - X_i}{h_n}\right) dx = 1$$

- Since $K(x) \geq 0$, then $\hat{f}(x) \geq 0$ for all x .

$\Rightarrow \hat{f}(x)$ is a probability density function

Next step: replace $K(x)$ by another probability density, maybe one giving more weight to points closer to x ?

KDE - Definition and Properties

Definition. KDE of f based on $X_1, \dots, X_N$ is

$$\hat{f}(x) = \frac{1}{nh_n} \sum_{i=1}^n K\left(\frac{X_i - x}{h_n}\right),$$

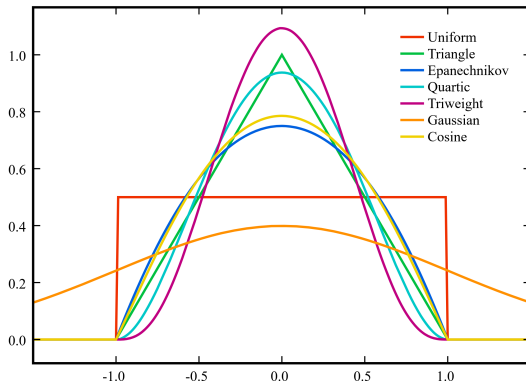
where the **kernel** $K(\cdot)$ satisfies:

- ① $K(x) \geq 0$ for all $x \in \mathbb{R}$
- ② $K(-x) = K(x)$ for all $x \in \mathbb{R}$
- ③ $\int_{\mathbb{R}} K(x) dx = 1$
- ④ $\lim_{|x| \rightarrow \infty} |x|K(x) = 0$
- ⑤ $\sup_x |K(x)| < \infty$

- $K(\cdot)$ is usually taken to be a density, and the assumptions
 - 1-3 hold if it is symmetric
 - 4 holds if it has a finite absolute moment
 - 5 holds if it is uniformly bounded
- if $h_n \rightarrow 0$ and $nh_n \rightarrow \infty$ (as $n \rightarrow \infty$), we have pointwise consistency
 - we will show this in a bit
 - also uniform consistency, but tricky to show

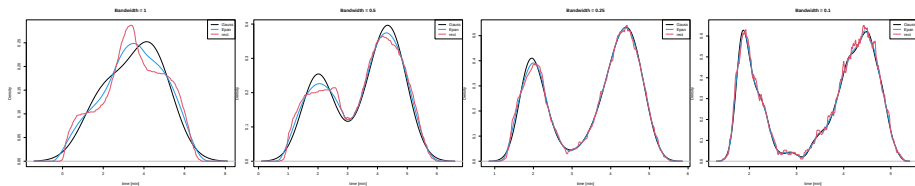
Common Kernels

Kernel Name	Formula
Epanechnikov	$K(x) \propto (1 - x^2)\mathbb{1}_{[x \leq 1]}$
Tricube (a.k.a. Triweight)	$K(x) \propto (1 - x ^3)^3\mathbb{1}_{[x \leq 1]}$
Gaussian	$K(x) \propto \exp(-x^2/2)$
...	...



Bandwidth $>$ Kernel

- While there is improvement when using non-rectangular kernels, the choice of the bandwidth is more important than that of the kernel
- A good choice is one that makes the estimate asymptotically converge quite rapidly in some well-chosen norm



Bias-Variance Trade-off

Goal: choose the tuning parameter h so that the mean squared error of the estimator is minimized:

$$\begin{aligned}\underbrace{\mathbb{E}[\{\hat{f}(x) - f(x)\}^2]}_{MSE\{\hat{f}(x)\}} &= \mathbb{E}[\{\hat{f}(x) - \mathbb{E}\hat{f}(x) + \mathbb{E}\hat{f}(x) - f(x)\}^2] \\ &= \underbrace{\{\mathbb{E}\hat{f}(x) - f(x)\}^2}_{bias^2} + \underbrace{\text{var}\{\hat{f}(x)\}}_{variance}\end{aligned}$$

Blackboard calculations (available in the lecture notes) give

$$\begin{aligned}\text{bias}\{\hat{f}(x)\} &= \frac{1}{2}h_n^2 f''(x) \int z^2 K(z) dz + o(h_n^2) \\ \text{var}\{\hat{f}(x)\} &= \frac{1}{nh_n} f(x) \int \{K(z)\}^2 dz + o\left(\frac{1}{nh_n}\right)\end{aligned}$$

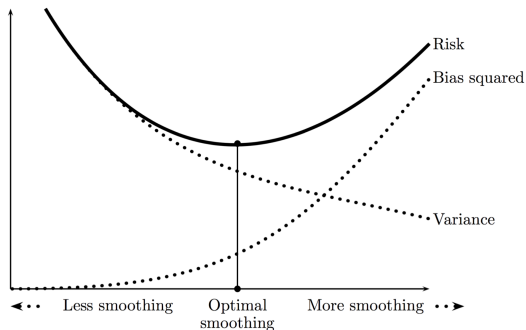
This shows consistency for $h_n \rightarrow 0$ and $nh_n \rightarrow \infty$ and the trade-off:

- small $h \Rightarrow$ small bias but large variance
- large $h \Rightarrow$ large bias but small variance

Bias-Variance Trade-off

The bias-variance trade-off is common when it comes to smoothing:

- In KDE, the smoothing is determined by the bandwidth
- Smoother estimates result in smaller variance but higher bias



Source: Wassermann (2006)

Optimal Bandwidth

Plugging this back in the MSE formula ignoring the *little-o* terms, differentiating the MSE w.r.t. h and setting it to zero leads to asymptotically optimal bandwidth choice:

$$h_{opt}(x) = n^{-1/5} \left(\frac{f(x) \int K(z)^2 dz}{[f''(x)]^2 [\int z^2 K(z) dz]^2} \right)^{1/5}$$

- $h_{opt}(x) \asymp n^{-1/5}$ ($h_{opt}(x)$ is of the order $n^{-1/5}$) and with this choice $MSE \asymp variance = \mathcal{O}(n^{-4/5})$
 - optimal non-parametric convergence rate
 - slower than the MSE of a MLE ($\mathcal{O}(n^{-1})$): price to pay for non-parametric approach
- $h_{opt}(x)$ is a local choice - depends on x . A global choice can be obtained by minimizing the MISE (to follow)

Optimal Kernel

The optimal bandwidth results in

$$MSE_{h_{opt}} = c(n, f) \left[\underbrace{\int x^2 K(z) dz \left\{ \int K^2(z) dz \right\}^2}_A \right]^{2/5},$$

where $c(n, f)$ is constant and depends only on n and f .

⇒ The optimal kernel is the one minimizing the term A .

It can easily be shown to be the Epanechnikov kernel!

Global Optimal Bandwidth

A common measure of performance of the estimator over all x is the Mean Integrated Squared Error (MISE):

$$\begin{aligned} MISE(\hat{f}) &= \mathbb{E} \int \{\hat{f}(x) - f(x)\}^2 dx \\ &= \int \mathbb{E}\{\hat{f}(x) - f(x)\}^2 dx = \int MSE\{\hat{f}(x)\} dx \end{aligned}$$

Minimizing the MISE yields the optimal bandwidth

$$\tilde{h}_{opt} = n^{-1/5} \left(\frac{\int K^2(z) dz}{\int \{f''(x)\}^2 dx [\int z^2 K(z) dz]^2} \right)^{1/5}$$

The resulting MISE is also of order $n^{-4/5}$

The Chicken and Egg problem

The theoretically optimal bandwidth $\tilde{h}_{opt} = n^{-1/5} (C(k) / \int \{f''(x)\}^2 dx)^{1/5}$ cannot be directly used as it depends on the unknown f . There are different approaches for the practical choice of h .

- **Reference method:** choose a parametric family for this formula
 - assume that f is the density of a $\mathcal{N}(\mu, \sigma^2)$ and then plug in its curvature $\frac{3}{8\sqrt{\pi}\sigma^5}$ into the formula of $\tilde{h}_{opt}$. This yields

$$\tilde{h}_{opt} = n^{-1/5} \sigma C(k)^{1/5} (8\sqrt{\pi}/3)^{1/5}$$

which when combined with a normal kernel gives the famous rule of thumb $\hat{h}_{opt} = (4/3)^{1/5} n^{-1/5} \hat{\sigma}$

- **Two-step method:** f in the formula is estimated non-parametrically by a pilot fit
 - estimate f'' by kernel estimate with pilot bandwidth
 - plug this estimate into $\tilde{h}_{opt}$ to estimate the optimal bandwidth in the kernel estimation of f

Section 2

Multivariate Density Estimation

Multivariate Density estimator

In practice, data are often multivariate

Consider n i.i.d. realizations of a d -dimensional random vector $\mathbf{X}_i = (X_{i,1}, \dots, X_{i,d})^\top$ from unknown F . We wish to estimate f , the density of F

The multivariate kernel density estimator is defined as

$$\hat{f}_n(\mathbf{x}) = \frac{1}{nh^d} \sum_{i=1}^n K\left(\frac{\mathbf{x} - \mathbf{X}_i}{h}\right),$$

where the kernel $K(\cdot)$ is a d -dimensional density

Multivariate Kernel

In practice, K is often chosen as

- the product of univariate kernels: $K(\mathbf{x}) = \prod_{j=1}^d K_0(x_j)$
- ellipsoidal kernel
 - multivariate normal density: $(2\pi)^{-d/2} \exp(-\mathbf{x}\mathbf{x}^\top/2)$

$\Rightarrow$ the matrix of bandwidths plays the role of the covariance-variance matrix

- multivariate Epanechnikov: $\frac{d+2}{2c_d}(1 - \mathbf{x}\mathbf{x}^\top)\mathbb{1}_{[-1,1]}(\mathbf{x}\mathbf{x}^\top)$, with c_d the volume of a d -dimensional unit ball ($c_1 = 1$, $c_2 = \pi$, $c_3 = 4\pi/3$)

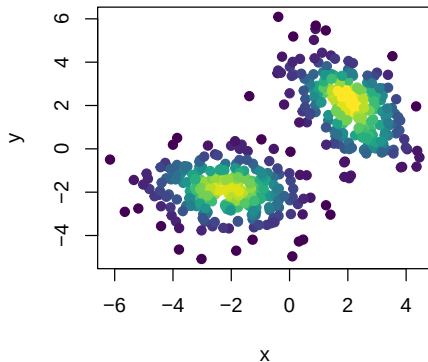
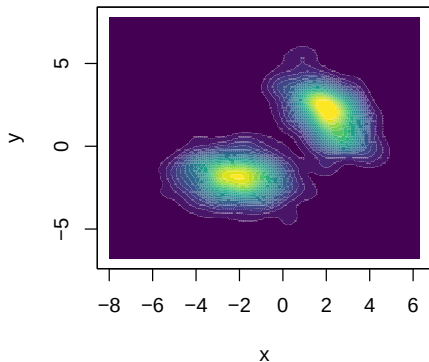
Degrees of smoothing are controlled by h and can be set different along the directions, i.e., under a product kernel, the KDE is

$$\hat{f}_n(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h_1} K_0\left(\frac{x_1 - X_{i,1}}{h_1}\right) \times \dots \times \frac{1}{h_d} K_0\left(\frac{x_d - X_{i,d}}{h_d}\right)$$

$\Rightarrow$ if margins are standardized (on the same scale), set $h = h_1 = \dots = h_d$

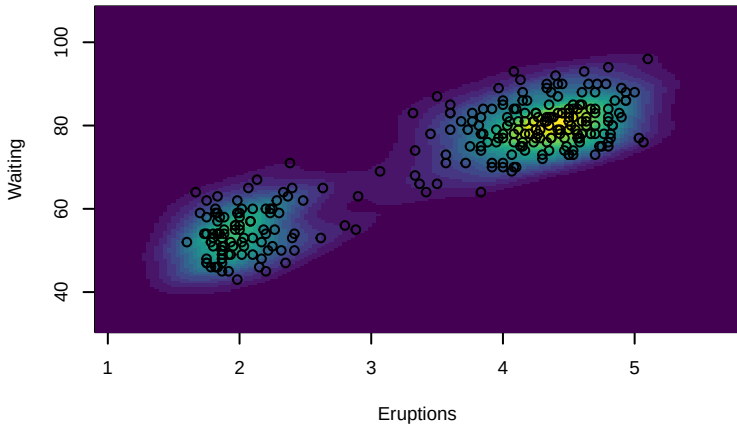
Multivariate KDE

- Mixture of bivariate normal



Multivariate KDE

- Faithful dataset



Curse of dimensionality

- KD estimation is typically restricted to $d = 2$
- Unless sample size is very large, neighbourhoods will be sparsely populated with data points (in higher dimensions)

For instance,

- If you have n data points uniformly distributed on the interval $[0, 1]$, how many data points are there in the interval $[0, 0.1]$?

Around $n/10$

- If you have n data points uniformly distributed on the 10-dimensional unit cube $[0, 1]^{10}$, how many are there in the cube $[0, 0.1]^{10}$?

Around $0.1^{10}n$

⇒ estimation gets harder very quickly as dimension increases

Curse of dimensionality

Under some smoothing conditions on f , the best possible MSE rate (the one obtained with optimal choice of bandwidth) is $O(n^{-4/(d+4)})$. That is, $MSE_{h_{opt}} \approx cn^{-4/(d+4)}$ and $n \approx (c/MSE_{h_{opt}})^{d/4}$

$\Rightarrow$ sample size grows exponentially with dimension

$n^{-4/(d+4)}$	$d = 1$	$d = 2$	$d = 5$
$n = 100$	0.025	0.046	0.129
$n = 1000$	0.004	0.010	0.046
$n = 10000$	6.3×10^{-4}	2.1×10^{-3}	1.6×10^{-2}

Thus, for $d = 5$, the rate with $n = 10000$ is the same than for $d = 2$ with 10 times less data ...

Summary - Overall

Motivation:

- 1 On Week 2, we introduced the histogram as a data exploratory tool and noticed its limitations
- 2 Histogram is a poor estimator of density, because it
 - is never smooth and requires a choice of *origin*
- 3 Today, we introduced naive KDE by generalizing histogram to its *origin*-free version
- 4 Then, we generalized naive KDE by allowing for better kernels
- 5 Now we have a decent nonparametric density estimation tool: KDE
 - in exploratory analysis, histograms often overlaid with KDEs

Main takeaways:

- 7 Asymptotic properties analyzed using Taylor expansions
 - suggest a way to choose *bandwidth*
 - the bias-variance trade-off made explicit
- 8 Multivariate extension works well in low dimensions

Assignment 2 and Exercise

Go to [Assignment 2](#) for details.

Go to [Exercise 2](#) for details.